# Guard cell size and pore aperture influence stomatal closure kinetics

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## notes

Kirschbaum et al. 1988 - read

Ochoa et al. (2024) - another study on correlation between stomatal density and size and gop

Tulva et al. (2025) - not super relevant for this paper.

Hsu et al. (2021) - t50 WT aribodopsis: 5.1 ± 0.4 min. Their VPD method “followed by □2.0 kPa VPD (20 to 32% relative air humidity) in most experiments, as indicated in the figures”. There was a lot in this paper. Basically, they don’t know exactly how VPD is sensed, but show that it’s independent of CO2 mechanism.

Reminder: Halliwell et al. 2025 paper has useful info on multiresponse models. I might use Van de Pol and Wright (2009) method for within versus between slope estimation. This is in the phylogenetic comparative example in the brms vignette too. Actually, this is hard to implement because I would have to calculate individual residuals for each light treatment combination and then means of those, etc.

Westoby et al (2023) (10.1111/1365-2745.14150) is similar to Halliwell. Maybe another way to think about it using the mutliresponse approach.

Some older references:

(this measured steady state responses, not kinetics) Morison J.I.L. Gifford R.M. 1983. Stomatal sensitivity to carbon dioxide and humidity. A comparison of two C3 and C4 grass species. Plant Physiol. 71:789–796.

(not really relevant) Talbott L.D. Rahveh E. Zeiger E. 2003. Relative humidity is a key factor in the acclimation of the stomatal response to CO2. J. Exp. Bot. 54:2141–2147.

hall_leaf_1976 - shows that gsw responds to VPD per se at constant LWP

Bunce (1997) - not very relevant

Blatt et al. (2026) - not sure if this is useful. Shows that guard cell walls stay hydrated even when airspace is subsaturated.

Buckley 2016 (10.1111/pce.12651) provides useful background on why hydroactive response takes time (ABA synthesis).

Merilo et al. (2018) say "The ABA signaling unit consists of cytosolic PYRABACTIN RESISTANCE1 (PYR1)/PYR1-LIKE (PYL)/ REGULATORY COMPONENTS OF ABA RECEPTORS (RCAR) receptors that, in the presence of ABA sequester type 2 protein phosphatases, lead to the activation of protein kinase OST1, which, in turn, activates the slow-type anion channel SLAC1 and triggers stomatal closure (Geiger et al., 2009; Lee et al., 2009; Ma et al., 2009; Park et al., 2009)" (these are steps after ABA synthesis)

Ivanoff 1928 - original wrong-way response citation? from Jalakas et al 2021 ("promoting a transient passive 'wrong-way' stomatal opening (Ivanoff, 1928; Mott & Franks, 2001; Buckley, 2019).")

FYI, t50 = "6.1 ± 0.32 min in Arabidopsis Col-0 wild-type (Merilo et al., 2018)" I also have read this: https://nph-onlinelibrary-wiley-com.ezproxy.library.wisc.edu/doi/full/10.1111/nph.17592

Need to check out these references "Although the plant hormone abscisic acid (ABA) is involved in stomatal closure in response to drought stress and reduced air humidity (Bunce, 1997; Roelfsema & Hedrich, 2005; Bauer et al., 2013), the existence of an ABA-independent pathway participating in low humidity-induced stomatal response has been suggested (Xie et al., 2006). Furthermore, stomata may also directly respond to leaf water status without relying on ABA production and signalling (see Buckley, 2005; Damour et al., 2010; Peak & Mott, 2011). However, both the metabolically mediated hydro-active feedback response of guard cell osmotic pressure to the water status of epidermal evaporative site (Buckley, 2005) and the stomatal response mechanism based on a near-equilibrium in water potential between the guard cells and the air in the stomatal pore (Peak & Mott, 2011) still embed the potential for ABA to be involved in air humidity-related stomatal behaviour."

Binstock et al. (2024) 10.1093/plphys/kiad560 - see if they have estimates of tau

This paper looked at epidermal cell : guard cell size in domesticated tomato and related to closure kinetics. See if I can do this? 10.1111/ppl.70543 (use Sack and Buckley estimate of epidermal cell size based on stomatal index, density, and guard cell size)

NOT THAT USEFUL - Tinoco-Ojanguren C, Pearcy RW. (1993) Stomatal dynamics and its importance to carbon gain in 2 rain forest Piper species. 1. VPD effects on the transient stomatal response to light flecks. Oecologia 94: 388–394

Merilo et al 2014 New Phyt - this paper decreased RH to 30-40%

Some OG citations "Changes in stomatal aperture are driven by changes in turgor pressure of guard cells (Heath 1938) due to the accumulation of ions and/or solutes (Imamura 1943; Fujino 1967; Outlaw 1983), which increases the osmotic potential and lowers the water potential of guard cells (Weyers and Meidner 1990; Willmer and Fricker 1996)." quote from Lawson et al (2010)

## Abstract

Guard cell size and stomatal density vary over X and X orders of magnitude among extant vascular land plants, yet the adaptive significance of much of this variation remains unclear. The evolution of guard cell size in particular is poorly understood and may even be constrained by nonadaptive features like genome size. One hypothesis is that natural selection may favor smaller guard cells to increase the rate at which stomatal conductance responds to fluctuating environmental conditions. A related hypothesis is that operational stomatal conductance ($g_{\mathrm{op}}$) is often $\approx 25\%$ of its theoretical maximum (gmax) because at this stomatal aperture, guard cell volume is poised to change rapidly with changes in turgor pressure. The support for both hypothesis is limited and mixed, in part because they have not been tested together, even though both guard cell size and the ratio of $g_{\mathrm{op}}$ to $g_{\mathrm{max}}$ ($f_{\mathrm{gmax}}$) should influence stomatal kinetics. We analyzed 2,126 estimates of stomatal closure kinetics in response to an abrupt increase in vapor pressure deficit (VPD) among 29 diverse wild tomato populations in the genus *Solanum*. Both smaller guard cell size and lower $f_{\mathrm{gmax}}$ were associated with faster stomatal closure kinetics, but at different levels of biological organization. Guard cell size explained X% of the variance in stomatal kinetics among populations, whereas gop:gmax explained X% of the variance among individuals within populations. [neither result would have been supported if analyzed in isolation - is this true?]. We conclude that guard cell size may respond to selection on stomatal kinetics over evolutionary time, but actual kinetics are modulated by a tradeoff between maximizing gop and minimizing the time required for stomatal closure. Putting these hypotheses together with stabilizing selection on gop may explain why stomatal size and density negatively covary among species.

## Introduction

In nature, change is the only constant. One way that vascular plants (tracheophytes) cope with change is by adjusting stomatal pore aperture to optimize the trade-off between carbon gain and water loss (1–3). In the absence of physical limits and energetic tradeoffs, natural selection would favor plants that could instantaneously adjust stomatal conductance ($g_{\mathrm{sw}}$) to perfectly track dynamic environmental conditions (4). In reality, stomatal responses take time, creating a lag between actual and optimal stomatal aperture. For example, many species use hydroactive feedback mechanisms to precisely control stomatal responses to changing vapor pressure deficit (VPD), the driving force for evaporation (5). Hydroactive stomatal

control requires sensing a change in VPD through as-yet-unknown mechanisms, signal transduction, and efflux of osmotic solutes to alter guard cell turgor more than the surrounding epidermal cells. Abscisic acid (ABA) dependent responses require several minutes to accumulate or synthesize hormone pools *de novo* (6–9); ABA-independent stomatal responses require changes in gene expression and protein synthesis that take time (10). Once anion channels are activated, the rate of change in stomatal aperture depends on the rate of change in guard cell turgor ($dP_g/dt$) and the sensitivity of aperture to change in turgor ($da_S/dP_g$). As we discuss below, these two factors ($dP_g/dt$ and $da_S/dP_g$) may be influenced by guard cell size and the ratio of operational to maximum stomatal conductance ($f_{gmax}$), respectively.

Over a century ago (11) observed that after an initial increase, leaf transpiration decreases in response to decreasing humidity. He correctly attributed changes in transpiration to changes in stomatal aperture and subsequent studies have confirmed that stomatal aperture responds to humidity (12) and, more specifically, transpiration rate (13, 14), which is the product of VPD and $g_{sw}$. The VPD is related to RH but also temperature. In angiosperms and some pteridophytes (15–17), a sudden increase in VPD, usually induced by lower RH at a constant temperature, causes a transient increase in $g_{sw}$, as Darwin observed. After a period of several minutes, stomata close until the leaf reaches a new steady-state $g_{sw}$. These phases of stomatal response to VPD are often referred to as the "wrong-way response" (WWR) and "right-way response" (RWR), respectively. The normative 'wrong' and 'right' appellations refer to the presumed adaptive significance. At steady state, $g_{sw}$ is negatively correlated with VPD, all else being equal, which is interpreted as an adaptive response to the marginal carbon cost of water (1, 18, 19). Models built around this assumption generally predict steady state $g_{sw}$ well, although the functional form and presumed fitness costs are areas of active research (20–24).

Stomatal responses are as much about the journey (kinetics) as they are about the destination (steady state). If stomatal response kinetics are too slow relative to the temporal grain size of environmental variability, then it would be futile for $g_{sw}$ to track environmental variation since the leaf will constantly lag the environment, perpetually fighting the last war. This may explain why $g_{sw}$ can be insensitive to short sun/shade flecks (25) and shaded leaves are likely primed to cope with the hydraulic demands of sudden increases in light, temperature, and VPD (26). If stomatal responses can track environmental variation, then minimizing the lag time can greatly improve resource-use efficiency (4). For example, optogenetic control of stomatal aperture in response to fluctuating light increases water-use efficiency in *Arabidopsis* (27). Therefore the rate of stomatal opening and closing is predicted to have a major effect on how natural selection optimizes stomatal control in variable environments (28).

Much of the research on stomatal kinetics focuses on light responses because light intensity is usually the most important, rapidly fluctuating environmental variable affecting photosynthesis. In nature, fluctuating light intensity simulateneously alters leaf temperature and VPD (e.g. 29). We focus on stomatal closure kinetic in response to an abrupt increased in for three reasons. First of all, the primary aims and *a priori* hypotheses of this study were not to study stomatal kinetic parameters and are described in (30). The idea to opportunistically use $g_{sw}$ response curves generated during this study to test hypotheses about which traits influence stomatal kinetics came after the data had been collected, but not yet analyzed for this purpose. Second, VPD responses exhibit a similar functional form as light-responses, suggesting similar underlying mechanisms (5). Factors that affect stomatal response kinetics to VPD likely generalize to other environmental variables. Third, humidity responses tend to

dominate over and are faster than other environmental responses (31, 32), but there is variation among species (33). Thus, factors that constrain maximum rates of stomatal closure are most likely to be observed in response to VPD. Finally, the ecological salience of VPD responses is increasing as climate change warms the atmosphere (34).

Fully describing stomatal responses to the environment is complex (9, 35, 36), but the dynamics of $g_{\mathrm{sw}}$ in response to an abrupt environmental change is well described by relatively simple phenomenological functions (37, 38). A four-parameter equation describes stomatal response kinetics to a step change in light intensity across dozens of plant species (39):

$$g_{\mathrm{sw,t}} = g_{\mathrm{f}} + (g_{\mathrm{i}} - g_{\mathrm{f}})e^{-\left(\frac{t}{\tau}\right)^{\lambda}}. \tag{1}$$

In this equation, $g_{\mathrm{sw,t}}$ is stomatal conductance at time $t$, $g_{\mathrm{i}}$ is the initial $g_{\mathrm{sw}}$ before the step change, $g_{\mathrm{f}}$ is the final $g_{\mathrm{sw}}$ after the step change, $\tau$ is a time constant that describes how quickly stomata respond to the step change, and $\lambda$ is a shape parameter that describes how the response rate changes over time. When $\lambda = 1$, the response rate is constant over time and $\tau$ represents the time required for $g_{\mathrm{sw}}$ to reach 36.8% of the way between initial and final values. When $\lambda > 1$, the response rate increases over time, and when $\lambda < 1$, the response rate decreases over time. (39) refer to $\lambda$ as a lag-time parameter because empirically $\lambda > 1$. We refer to stomatal kinetic parameters $\tau$ (time-constant) and $\lambda$ (lag-time) henceforth. The first goal of this study is to evaluate whether kinetics of the RWR after a step change in VPD are adequately described by Equation 1. If so, it suggests that some fundamentally similar mechanisms are shared between light and hydroactive stomatal responses (40).

The main goal of this study is to test hypotheses about which stomatal anatomical traits affect kinetic parameters $\tau$ and $\lambda$. We consider two interconnected hypotheses that have thus far been treated in isolation. The first hypothesis is that smaller guard cells open and close faster because of their intrinsically greater surface area to volume ratio (39, 41–43). For approximately cylindrical cell geometries, like that found in guard cells, surface area increases linearly with radius, whereas volume increases in proportion to the radius squared. Consequently, larger guard cells will require more time, all else being equal. to achieve a given change in turgor pressure and, hence, stomatal pore aperture. A plant with large guard cells can partially compensate for this geometric constraint by increasing the activity and/or density of ion and solute transporters in the guard cell plasma membrane (44–46). However, this may be limited by the available membrane area and/or energetic costs of maintaining high transporter densities (37). Empirical support for the hypothesis that smaller guard cells are faster is mixed and has primarily come from step changes in light (38, 42, 47–49). Meta-analysis of dozens of species suggests that stomatal size alone is weakly related to kinetics and modulated by whether guard cells are kidney or dumbbell shaped (39). The relationship between guard cell size and stomatal response kinetics to leaf water potential (50) and VPD (51, 52) is less well studied, but similarly equivocal.

## MARKER

One factor that might complicate the relationship between size and speed is aperture. Stomatal aperture can vary from near 0 when guard cell turgor pressure is low and asymptotically approach $g_{\mathrm{max}}$ as guard

cell turgor pressure approaches $\infty$ (Figure 1X). The relationship between guard cell turgor pressure and stomatal aperture is nonlinear (53) because cell walls become less elastic at greater turgor (54). The nonlinear relationship between guard cell turgor and pore aperture implies that if the rate of change in turgor change is constant ($\frac{\mathrm{d}P}{\mathrm{d}t} = C$), the rate of change in aperture, and hence stomatal conductance, will vary depending on the initial aperture. When initial aperture is high, stomatal conductance will respond more slowly than when initial aperture is low. Since we generally do not observe individual stomatal aperture, extending this idea to macroscale phenomena requires scaling by stomatal density. We will use the term "fraction of anatomical maximum stomatal conductance", symbolized as $f_{\mathrm{gmax}}$ (elsewhere this is denoted as $g_{\mathrm{ratio}}$ by (55)). This value should be proportional to the average aperture divided by its maximum aperture for any arbitrary stomatal density [could a figure show how this interacts with stomatal density?]. Leaves operating at $f_{\mathrm{gmax}}$ closer to unity will, all else being equal, will be slower to respond than leaves operating with a $f_{\mathrm{gmax}}$ close to zero, independent of stomatal density. (56) hypothesized that selection on stomatal density would maintain typical operational stomatal conductance (gop) relative to gmax at a sufficiently low value that stomatal conductance would be sensitive to relatively small changes in guard cell turgor pressure. Consistent will this hypothesis, gop:gmax is often near 0.25 (57, 58), well below $f_{\mathrm{gmax}} = 1$, and in a range where $g_{\mathrm{sw}}$ would be responsive to small changes in guard cell turgor pressure.

Putting these two hypotheses together, we predict that both guard cell size and fgmax will influence stomatal kinetics, but possibly at different scales of biological organization. Guard cell size tends to vary less than stomatal density or aperture at many biological scales. For example, on a mature leaf, guard cell size is essentially fixed other than small changes in volume caused by turgor pressure (cite). Compared to stomatal density, guard cell size is also less developmentally plastic varies less genetically within species (cites). If there is relatively little plastic or genetic variation among individuals within a species, then most of the variance in stomatal kinetics within species cannot be explained by variation in stomatal size. In contrast, $f_{\mathrm{gmax}}$varies immensely within and among individuals because stomatal aperture responds dynamically over the day and in response to environmental variation. Within a single leaf, $f_{\mathrm{gmax}}$will change over the course of a day in response to light signalling, change in VPD, and starch accumulation (or sink strength? cite). During the life a leaf, $f_{\mathrm{gmax}}$will change in response to long-term stresses such as drought. Because guard cell size and aperture typically vary at different levels of biological organization, we predict that guard cell size likely explains more variation in stomatal kinetics among species, whereas $f_{\mathrm{gmax}}$will explain more variation within species. Within species variation can arise from either genetic or environmental differences between individuals. Neither of these traits along will determine all or even most of the variance in stomatal kinetics, which is a complex response to many internal and external signals (cites).

Here we extend recent advances in phylogenetic comparative methods (59–61) to test these predictions using data on stomatal closure kinetics in response to an abrupt increase in VPD among 29 diverse wild tomato populations in the genus *Solanum*. We leveraged natural variation in guard cell size among and within species to test whether leaves with smaller guard cells close faster. We induced variation in fgmax through a combination of growth and measurement light intensity. Growth light intensity caused developmental plasticity in stomatal density that, when crossed factorially with measurement light intensity, resulted in variation in fgmax. By measuring multiple individuals from multiple treatments

across multiple species enabled us to estimate the effect of guard cell size and $f_{\mathrm{gmax}}$ on stomatal kinetics and partition their significance within and among species.

Four main questions (where does this go?)

1. Are VPD kinetics adequately characterized by the same functional form as light responses?
2. Are $l_{\mathrm{gc}}$ and $f_{\mathrm{gmax}}$ positively associated with $\tau$?
3. At what biological levels of organization do $l_{\mathrm{gc}}$ and $f_{\mathrm{gmax}}$ influence $\tau$?
4. Are abaxial-only stomatal responses slower than whole leaf stomatal responses?

Before analyzing the data, we planned to test two hypotheses based on a recent metaanalysis of stomatal responses to light (39). We hypothesized that 1) guard cell size would influence $\tau$ and 2) that $\tau$ would be greater on the abaxial-only leaves. We planned to compare relationships within and among populations using phylogenetic comparative methods. We made the decision to test for an association between $f_{\mathrm{gmax}}$ and $\tau$ after exploratory analyses revealed a potential association. Adding hypotheses after some results have been analyzed, also known as "Hypothesizing After Results Known" (HARKing), can lead to reports of spurious associations that are not repeatable in later studies (62, 63). Therefore, our conclusions about the effect of $f_{\mathrm{gmax}}$ on stomatal kinetics should be interpreted with caution until they are replicated.

[not sure where this goes] many, large - slow because of large size, but fast because of low gop few, small - slow because of high gop, but fast because of small size some, moderate - best of both worlds? IDEA FOR FIGURE - use empirical estimates to predict tau (z-axis) for a given optimal gs as a function of

Is there a connection here to size-density scaling? It's an empirical regularity, not a law of nature that higher gmax is associated with smaller stomata. This is an observation that requires explanation, not a presumption of truth.

What's never been done before is to put these two ideas together. This means that weak relationships between size and speed might be because people have not controlled for aperture (gop:gmax ratio).

Factors cannot also manifest at different scales of biological organization. Here we focus on two levels: variation among species and variation among individuals within species. To consider it abstractly, if factors A and B influnce trait C according to some mechanistic model, but factor A is typically fixed within species but varies among species, then B will explain much of the variation in C within species. But if A varies among species, then A will explain much of the variation in C among species. We hypothesize that stomatal size is relatively constant within species, but varies among species. In contrast, aperture (gop:gmax) varies because of mismatches between leaf anatomy and gopt.

And that smaller stomata are faster.

These are related because the benefit of small stomata goes away if gop is really high.

hooks:

Haworth et al (2018) paper arguing faster adaxial kinetics (I think). Check other cites in Woning and Horak. It would be good to mention this " Mott (2007) reported that stomatal responses to humidity were

independent on the two surfaces of amphistomatous leaves of Vicia faba and Xanthium strumarium: when humidity was changed at one surface only, stomata on that surface responded but stomata on the other surface did not." (quote from Buckley et al 2011). FYI, tomatoes have BSEs (heterobaric)

Stomatal kinetics are important for optimizing response to rapid fluctuations in light, leaf temperture, and VPD

curve shape does not match what you'd expect for constant alpha term

Anatomy possibly matters: shape and size

Adaxial stomata might close faster (why?)

What we did:
1. model stomatal close with hysteresis (can it explain overshoot and wrong-way response?
2. compare tau in tomatoes to that in rice
3. test for effect of size
4. test if adaxial+abaxial is faster than abaxial alone

## Materials and Methods

### Populations and phylogeny

We analyzed stomatal responses to changing humidity among 29 ecologically diverse populations of wild tomato, including representatives of all described species of *Solanum* sect. *Lycopersicon* and sect. *Lycopersicoides* (64) and the cultivated tomato *S. lycopersicum* var. *lycopersicum* (Table 1), as described in (30). These populations were selected because they encompass the breadth of climatic variation in the wild tomato clade (65). In one case, we substituted a congeneric population of the species (*S. galapagense*) because of difficulty growing the focal population and we also added a population of *S. pennellii*. Due to constraints on growth space and time, we spread out measurements over -Inf weeks from Inf to -Inf. Replicates within population were evenly spread out over this period to prevent confounding of temporal variation in growth conditions with variation among populations. To estimate phylogenetic effects, we used the RAxML whole-transcriptome concatenated phylogeny based on 2,745 100-kb genomic windows from (65). Two of our populations were not in this tree. We used accession LA1044 in place of LA3909, two populations of *S. galapagense*; LA0750 was added as sister to LA0716, two closely related populations of *S. pennellii*. The node separating LA0750 from LA0716 was placed half-way between the next deepest node.

Table 1: Accession information of *Solanum* populations used in this study. The species name, accession number, collection latitude, longitude, and elevation. TGRC: Tomato Genetics Resource Center.

| Species | TGRC accession | Latitude | Longitude | Elevation (mas) |
|---|---|---|---|---|
| *S. arcanum* | LA2172 | -6.008 | -78.858 | 662 |
| *S. cheesmaniae* | LA0429 | -0.644 | -90.329 | 800 |
| *S. cheesmaniae* | LA3124 | -0.804 | -90.042 | 1 |
| *S. chilense* | LA1782 | -15.267 | -74.633 | 1000 |
| *S. chilense* | LA4117A | -22.907 | -67.941 | 3540 |
| *S. chmielewskii* | LA1028 | -13.883 | -73.017 | 3000 |
| *S. chmielewskii* | LA1316 | -13.400 | -73.906 | 2920 |
| *S. corneliomulleri* | LA0107 | -13.117 | -76.383 | 60 |
| *S. corneliomulleri* | LA0444 | -13.433 | -76.133 | 100 |
| *S. galapagense* | LA0436 | -0.953 | -90.978 | 40 |
| *S. galapagense* | LA1044 | -0.284 | -90.548 | 0 |
| *S. habrochaites* | LA0407 | -2.181 | -79.884 | 70 |
| *S. habrochaites* | LA1777 | -9.550 | -77.700 | 3216 |
| *S. huaylasense* | LA1358 | -9.533 | -77.967 | 750 |
| *S. huaylasense* | LA1360 | -9.546 | -77.929 | 1490 |
| *S. huaylasense* | LA1364 | -10.133 | -77.383 | 2920 |
| *S. lycopersicoides* | LA2951 | -19.317 | -69.450 | 2200 |
| *S. lycopersicoides* | LA4126 | -19.287 | -69.396 | 3120 |
| *S. neorickii* | LA1322 | -13.483 | -72.442 | 2380 |
| *S. neorickii* | LA2133 | -3.400 | -79.183 | 1980 |
| *S. pennellii* | LA0716 | -16.225 | -73.617 | 50 |
| *S. pennellii* | LA0750 | -14.775 | -75.034 | 550 |
| *S. pennellii* | LA3778 | -14.775 | -75.034 | 616 |
| *S. peruvianum* | LA2744 | -18.550 | -70.150 | 400 |
| *S. peruvianum* | LA2964 | -18.028 | -70.835 | 75 |
| *S. pimpinellifolium* | LA1269 | -11.483 | -77.075 | 400 |
| *S. pimpinellifolium* | LA1589 | -8.433 | -78.817 | 30 |
| *S. pimpinellifolium* | LA2933 | -1.442 | -80.562 | 375 |
| *S. sitiens* | LA4116 | -22.159 | -68.782 | 2960 |

**Plant growth conditions and light treatments**

A thorough description of plant growth conditions can be found (30), therefore we summarize the most important information here. Seeds provided by the Tomato Genetics Resource Center germinated on

moist paper in plastic boxes after soaking for 30-60 minutes in a 50% (volume per volume) solution of household bleach and water, followed by a thorough rinse. We transferred seedlings to cell-pack flats containing Pro-Mix BX potting mix (Premier Tech, Rivière-du-Loup, Quebec, Canada) once cotyledons fully emerged, typically within 1-2 weeks of sowing. We grew seeds and seedlings for both sun and shade treatments under the same environmental conditions (12:12 h, 24.3:21.7 °C, 49.6:58.4 RH day:night cycle). LED light provided PPFD $= 267\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ (Fluence RAZRx, Austin, Texas, USA) during the germination and seedling stages.

Immediately prior to starting growth light intensity treatments (sun and shade), we transplanted seedlings to 3.78 L plastic pots containing 60% Pro-Mix BX potting mix, 20% coral sand (Pro-Pak, Honolulu, Hawai‘i, USA), and 20% cinders (Niu Nursery, Honolulu, Hawai‘i, USA) withslow release NPK fertilizer following manufacturer instructions (Osmocote Smart-Release Plant Food Flower & Vegetable, The Scotts Company, Marysville, Ohio, USA). Percentage composition is on a volume basis. We watered to field capacity three times per week to prevent drought stress. Seedlings were randomly assigned in alternating order within population to the sun or shade treatment during transplanting. The average daytime PPFD was $761\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ and $115\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ for sun and shade treatments, respectively.

**Humidity response curves**

We selected a fully expanded, unshaded leaf at least six leaves above the cotyledons during early vegetative growth. This typically meant that plants had grown in light treatments for $\approx 4$ weeks, ensuring they had time to sense and respond developmentally to the light intensity of the treatment rather than the seedling conditions (66). See (30) for further detail.

To measure stomatal responses to a step change in humidity, we first acclimated the focal leaf to high humidity, rapidly decreased the humidity, and logged $g_{\text{sw}}$through time. We refer to this as a humidity-response curve. We measured humidity-response curves using a portable infrared gas analyzer (LI-6800PF, LI-COR Biosciences, Lincoln, Nebraska, USA). Light-acclimated plants were placed under LEDs dimmed to match their light treatment during gas exchange measurements. We estimated $g_{\text{sw}}$at ambient $CO_2$ ($C_{\text{a}} = 415\,\mu\text{mol}\,\text{mol}^{-1}$) and $T_{\text{leaf}} = 25.0\,°\text{C}$.

We collected four humidity-response curves per leaf, an amphi (untreated) curve and a pseudohypo (treated) curve at high light-intensity (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 97.8:2.24 red:blue) and low light-intensity (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$; 87.0:13.0 red:blue). We always measured high light-intensity curves first because photosynthetic downregulation is faster than upregulation in these species. Pseudohypostomatous leaves are the same as the amphistomatous leaves except gas exchange through the upper (adaxial) surface blocked by a neutral density plastic (propafilm). Comparing amphi and pseudohypo curves on the same leaf enables us to test whether stomatal kinetics are different for ab- and adaxial stomata on the same leaf. To compensate for reduced transmission, we increased incident PPFD for pseudohypo leaves by a factor 1/0.91, the inverse of the measured transmissivity of the propafilm. We also set the stomatal conductance ratio, for purposes of calculating boundary layer conductance, to 0 for pseudohypo leaves following manufacturer directions.

Table 2

| | Measurement light intensity | | | | |
|---|---|---|---|---|---|
| | Low | | High | | |
| Growth light intensity | amphi | pseudohypo | amphi | pseudohypo | Total |
| shade | 275 | 275 | 270 | 266 | 1,086 |
| sun | 245 | 245 | 275 | 275 | 1,040 |
| Total | 520 | 520 | 545 | 541 | 2,126 |

To control for order effects, we alternated between starting with amphi or pseudohypo leaf measurements. We made measurements over two days. On the first day, we measured high and low light-intensity curves for either amphi or pseudohypo leaves; on the second day, we measured high and low light-intensity curves on the other leaf type. The irradiance of the light source in the pseudohypo leaf was higher because the propafilm reduces transmission.

In all cases, we acclimated the focal leaf to high light (PPFD $= 2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$) and high relative humidity (RH $= 70\%$) until $g_{\text{sw}}$ reached its maximum. After that, we scrubbed water vapor from the incoming air using silica desiccant as quickly as possible, resulting in a water vapor concentration in the incoming air to be on average $0.36\,\text{mmol}\,\text{mol}^{-1}$. This treatment induced rapid stomatal closure, but it did not standardize chamber air humidity because water vapor was introduced by leaf transpiration. HOWEVER - NO RELATIONSHIP BETWEEN FINAL GSW AND TAU (fill in details). This treatment was designed to elicit a rapid stomatal response and not necessarily mimic natural environmental variability. All other environmental conditions in the leaf chamber remained the same. We started logging data after the transient "wrong-way response" (citation) and continued logging data until $g_{\text{sw}}$ reached its nadir. We then acclimated the leaf to low light (PPFD $= 150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$) and RH $= 70\%$ before inducing stomatal closure with low RH and logging data as described above.

Prior to analysis, we removed unreliable and unusable data points, removed points not at instrument steady state, removed hysteretic portion of each curves at low $g_{\text{sw}}$, removed outliers within each curve, removed replicates with no overlap between amphi and pseudohypo curves from the same leaf, and thinned redundant data points within each curve. The rationale and procedure for each cleaning step is described in (30). Additionally, we excluded stomatal kinetic parameter estimates from 4 curves where $\tau$ was estimated to be greater than 1097 s, greatly exceeding the typical estimates (see Results). After these steps, the total number of humidity response curves analyzed is given in Table 2. The median number of curves per population per treatment was 10 and the median number of points per curve was 26. We extracted the the posterior distribution for $\tau$ and $\lambda$ from each fitted curve to calculate the parameter estimate (median) and uncertainty (standard deviation) for subsequent analyses.

### Stomatal anatomical traits $l_{gc}$ and $f_{gmax}$

We estimated the stomatal density and $l_{gc}$ on ab- and adaxial surfaces from all leaves used for humidity response curves. We used standard methods describe fully in (30). For each surface we calculated the anatomical maximum stomatal conductance ($g_{max}$) at a reference leaf temperature of 25 °C following (67) as:

$$g_{max} = bmdl_{gc}/\sqrt{2}.$$

The biophysical and morphological constants $b$ and $m$ are:

$$b = \frac{D_{wv}}{v}, \text{ and}$$
$$m = \frac{\pi c^2}{j^{0.5}(4hj + \pi c)},$$

where $D_{wv}$ is the diffusion coefficient of water vapor in air and $v$ is by the kinematic viscosity of dry air. We assumed $D_{wv} = 2.49 \times 10^{-5}\ \mathrm{m^2\ s^{-1}}$ and $v = 2.24 \times 10^{-2}\ \mathrm{m^3\ mol^{-1}}$ (68). For kidney-shaped guard cells like those in wild tomatoes, $c = h = j = 0.5$. The $g_{max,ratio}$ is the $g_{max}$ of the adaxial surface divided by the sum total $g_{max}$ of both surfaces. To estimate $f_{gmax}$, we divided the initial $g_{sw}$ estimated from the parameter $g_i$ in Equation 1 from each humidity response curve the anatomical $g_{max}$ for that leaf. The estimated $g_i$ was very close to the maximum $g_{sw}$ observed near the beginning of each curve (Figure S2), suggesting that the initial $g_{sw}$ was a good proxy for operational stomatal conductance at the time of the step change in humidity. We used the anatomical $g_{max}$ for the surface(s) measured in each curve to calculate $f_{gmax}$. For amphi curves, we used the sum of adaxial and abaxial $g_{max}$; for pseudohypo curves, we used only the abaxial $g_{max}$.

### Hypothesis generation and testing

I MOVED SOME OF THIS TO INTRO. NEED TO RENAME SUBSECTION, POSSIBLY RESTRUCTURE SOME

We estimated $\tau$ and $\lambda$ for each response curve by fitting Equation 1 to a time-series of $g_{sw}$ values using Bayesian nonlinear regression. All models were fit using a Bayesian HMC sampling in the probabilistic programming language *Stan* (69) using the *R* package **brms** version 2.23.0 (70). We used *CmdStan* version 2.38.0 and **cmdstanr** version 0.9.0 (71) to interface with *R* version 4.5.2 (72). We sampled 1000 post-warmup draws from the posterior distribution on a single chain. We adjusted the number sampling iterations and thinning interval for each curve so that the Gelman-Rubin convergence statistic ($\hat{R}$) (73) was less than 1.05 and the bulk effective sample size (ESS) was greater than 400 for all model parameters, and there were fewer than 10 divergent transitions. We assessed model fit using Bayesian $R^2$ (74) implemented in **brms**.

MOVE/REVISE? We next used estimates of $\tau$ from each curve in multiresponse Bayesian mixed effects models with phylogenetically structured random effects to answer the following core questions:

We used a combination of model selection and parameter estimation to evaluate answer the three main questions (see Introduction). All statistical models included fixed effects of manipulative treatments: growth light intensity (sun and shade); measurement light intensity (high and low); and curve type (amphistomatous and pseudohypostomatous). All models also included population as both a phylogenetically structured random effect and a non-phylogenetic random effect following (61). Prior to fitting, the phylogenetic (co)variance matrix was rescaled as a correlation matrix (75). For traits measured multiple times within the same leaf ($f_{\mathrm{gmax}}$, $\lambda$, $\tau$) we included a random effect of individual to account for repeated measures. We accounted for uncertainty in $\tau$ and $\lambda$ using the standard deviation of the posterior distribution of each estimate (see 'Humidity response curves' for details). We modeled residual (co)variance, a combination among-individual (co)variance and measurement error, as a multivariate Student $t$-distribution, which is more robust to extreme values than the multivariate normal distribution (76). To account for heteroskedasticity, bounded measurements, and linearize relationships, $l_{\mathrm{gc}}$, $\lambda$, and $\tau$ were log-transformed prior to analysis whereas $f_{\mathrm{gmax}}$ was logit-transformed. All models were fit with *Stan* following the same methods described above for fitting Equation 1 to the humidity response curves.

To answer questions 1 and 2 above, we analyzed three nonmutually exclusive paths through which $f_{\mathrm{gmax}}$ and $l_{\mathrm{gc}}$ could influence stomatal kinetic parameters $\lambda$ and $\tau$. First, we inferred phylogenetic effects by estimating the phylogenetic correlation among populations between anatomical ($f_{\mathrm{gmax}}$ and $l_{\mathrm{gc}}$) and kinetic ($\lambda$ and $\tau$) variables. To test for possible causation, we used the inferred phylogenetic trait covariance matrix to estimate partial correlations between anatomical and kinetic variables (61). We interpret a significant partial correlation between variables as a plausible causal hypothesis. Second, we inferred individual-level effects of $f_{\mathrm{gmax}}$ and $l_{\mathrm{gc}}$ on kinetic parameters using linear multiple regression that simultaneously accounts for phylogenetic and treatment effects. We used the leave-one-out cross-validation information criterion (LOOIC) to compare the fit of models using the *R* package **loo** version 2.9.0 (77) to calculate LOOIC values. We compared models with all permutations of $f_{\mathrm{gmax}}$ and $l_{\mathrm{gc}}$ influencing $\lambda$ and $\tau$. We considered all models within two $\Delta$LOOIC standard errors of the minimum LOOIC (Table 6) to be in the set of plausible models to generate posterior predictions for hypothesis testing. Within each plausible model, we assessed whether an effect was significantly different from zero based on the 95% confidence intervals. Third, we used causal mediation analysis (78) to test whether plasticity in $\tau$ was mediated by plasticity in $f_{\mathrm{gmax}}$ or $l_{\mathrm{gc}}$ in response to measurement light intensity, growth light intensity, and curve-type treatments. We inferred the direct effect of treatments on $\tau$ from model coefficients. We estimated the mediated effect of treatments using the product of the effect of treatment on mediator ($f_{\mathrm{gmax}}$ or $l_{\mathrm{gc}}$) and the effect of the mediator on $\tau$. We determined significant direct and mediator effects based on whether the 95% confidence intervals of the effects were different from zero.

ADD TO THIS PARAPGRAPH, To compare the strength and statistical significance of explanatory variables, we estimated the effect size as the point estimate (median of the posterior) divided the standard error (standard deviation of the posterior) for each effect.

### Variance decomposition and heritability (somewhere)

We decomposed the variance in log-transformed response variables ($\lambda$, $tau$, $f_{\text{gmax}}$, $l_{\text{gc}}$) into phylogenetic, population, between-individual, and within-individual levels. The phylogenetic variance component quantifies trait evolution between populations congruent with the phylogenetic relationships, whereas the population component quantifies variance among populations independent of phylogenetic relationships (75, 79). The between- and within-individual variance components quantify variance among individuals within the same population and treatment. The within-individual component was estimated from repeated measures of the same leaf at different light intensities. It was not possible to estimate within-individual variance in guard cell length with our study design because we only measured average guard cell length once per leaf. We estimated the between-individual variance from the residual variance, which also includes measurement error. The phylogenetic heritability is equivalent to the phylogenetic variance component and is estimated following (79); (75) as:

$$h^2_{\text{phy}} = \frac{\sigma^2_{\text{phy}}}{\sigma^2_{\text{phy}} + \sigma^2_{\text{pop}} + \sigma^2_{\text{between}} + \sigma^2_{\text{within}}},$$

where $\sigma^2_{\text{phy}}$, $\sigma^2_{\text{pop}}$, $\sigma^2_{\text{between}}$, and $\sigma^2_{\text{within}}$ are the phylogenetic, population, between-individual, and within-individual variance components, respectively.

### Model comparison (somewhere)

add info

## Results

### The time-constant $\tau$ characterizes stomatal kinetics

Stomatal conductance ($g_{\text{sw}}$) decreased rapidly in response to a step change in humidity. The model in Equation 1 fit the data extremely well (Figure S1). The mean and minimum Bayesian $R^2$ across all curves was 0.9994 and 0.9802, respectively. Consider a typical leaf in the experiment characterized by the median time constant ($\tau$) and lag time ($\lambda$) among wild tomato accessions in all treatments. After humidity decreased and the transient wrong way response elapsed, it took 121 s for $g_{\text{sw}}$to decrease half-way ($t_{50}$) from its initial ($g_{\text{i}}$) to final (($g_{\text{f}}$) steady state value. In terms of the kinetic model parameters, most variation among leaves was due to difference in the time constant rather than the lag time. In the previous example, increasing $\lambda$ from its median to its maximum estimated value among accessions increased the $t_{50}$ by only 8.2 s. In contrast, increasing $\tau$ from its median to maximum value increased $t_{50}$ by 129 s. Hence, we primarily focus on treatments and anatomical parameters affect $\tau$.

### Effects of light intensity treatments on guard cell length and aperture

On average, guard cell length increased by 12.1% [95% CI: 11.2 to 13.1%] in the sun treatment (Figure 1a; Table 5). The average $l_{gc}$ of pseudohypo leaves was $-5.9\%$ [95% CI: $-6.7$ to $-5.1\%$] greater than amphi leaves (Figure 1a; Table 5). This was not because of plasticity because we amphi and pseudohypo curves on the same leaf. Rather, this was a composition effect. Adaxial stomata tended to be slightly smaller than abaxial stomata, so the average $l_{gc}$ in pseudohypo leaves was slightly higher than in amphi leaves because the former only included abaxial stomata. High measurement light intensity increased $f_{gmax}$by an average of $0.84$ logit units [95% CI: $0.82$ to $0.87$] (Figure 1b; Table 5). For example, if $f_{gmax} = 0.1$ in low light, the typical $f_{gmax}$ for the same leaf acclimated to high light was 0.21. Sun leaves had on average $-0.28$ logit units [95% CI: $-0.35$ to $-0.21$] lower $f_{gmax}$ than shade-grown plants (Figure 1b; Table 5). Treated pseudohypo leaves had on average $0.21$ logit units [95% CI: $0.18$ to $0.24$] higher $f_{gmax}$ than untreated, amphistomatous leaves (Figure 1b; Table 5). This indicates that stomatal aperture was more open in leaves where gas exchange through adaxial stomata was blocked.

### Effects of light intensity treatments on $\tau$

Growth light intensity did not significantly affect $\tau$ Figure 2. In sun plants, $\tau$ was on average 5.64% [95% CI: 1.03 to 10.6%] lower than in shade plants after accounting for other explanatory variables. Within the same leaf, increasing the measurement light intensity prior to the humidity step change from $150\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ to $2000\,\mu\text{mol}\,\text{m}^{-2}\,\text{s}^{-1}$ significantly increased $\tau$ by 18.3% [95% CI: 11.8 to 25%] Figure 2. The time lag $\lambda$ responded to growth, but not measurement light intensity after accounting for other explanatory variables Figure 2. In sun plants, $\tau$ was on average 12.2% [95% CI: 10.2 to 14.2%] greater than in shade plants. Higher measurement light decreased increased $\lambda$ slightly by 1.2% [95% CI: -1.07 to 3.56%].

- table of model estimates for tau and lambda

| Species | mean | sd |
|---|---|---|
| setosa | 5.006 | 0.3524897 |
| versicolor | 5.936 | 0.5161711 |
| virginica | 6.588 | 0.6358796 |

### Variance decomposition and phylogenetic heritability

Among the four traits in our model ($f_{gmax}$, $l_{gc}$, $\lambda$, and $\tau$), the preponderance of the variance was typically within or between individuals after accounting for treatment effects Figure 3, Table 4. The exception to this pattern was $l_{gc}$, for which the phylogenetic component (aka phylogenetic heritability) was larger Figure 3. The population component independent of phylogeny explained little of the trait variation. The between-individual variance was usually high, but this component confounds true between-individual variance with measurement error, so should be interpreted with caution.

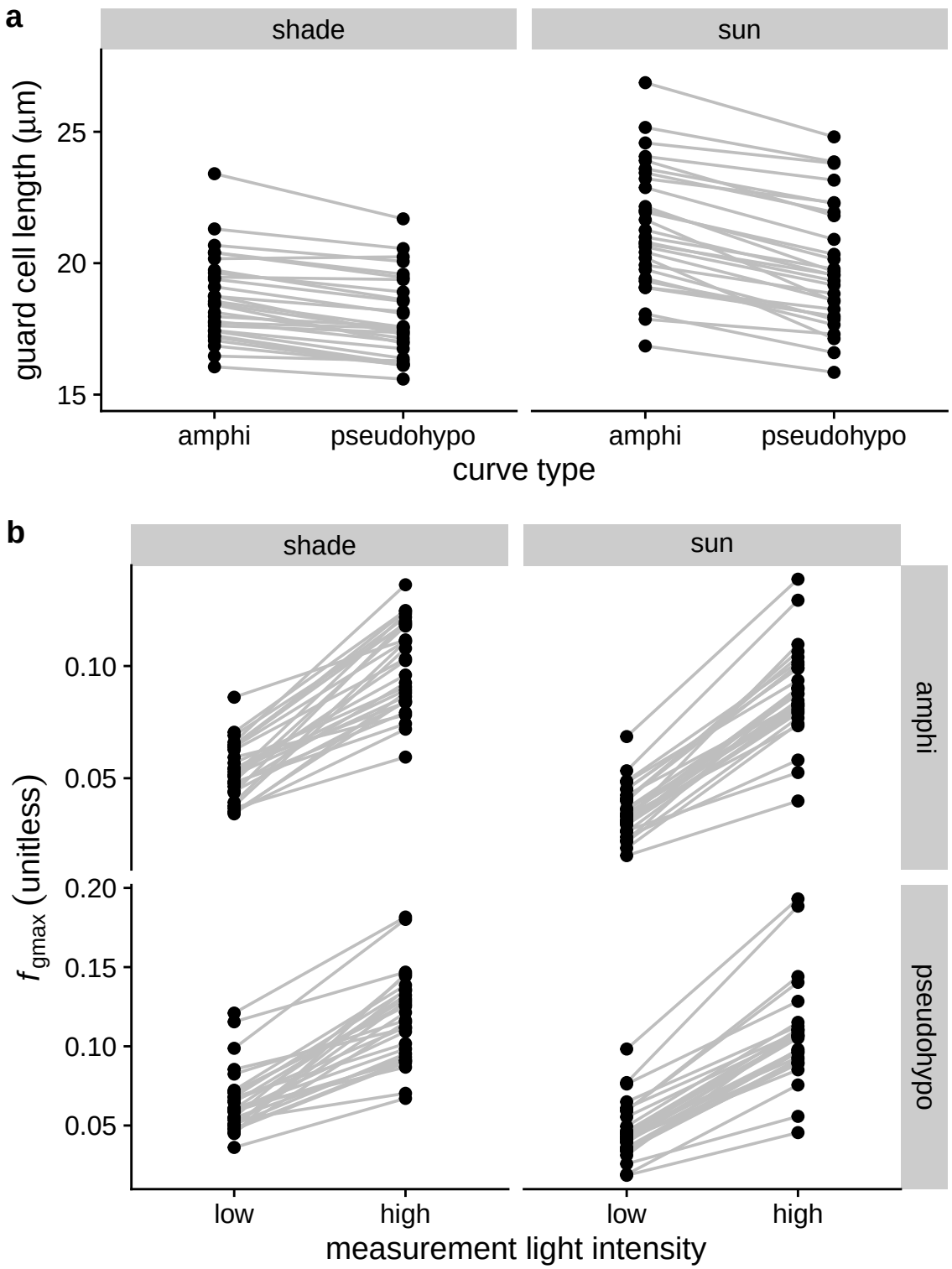


Figure 1: **Guard cell length ($l_{gc}$) and stomatal conductance as a fraction a maximum conductance ($f_{gmax}$) vary among populations and treatments.** In both panels, black points are mean trait value per population within a particular treatment. Gray lines connect values within populations across treatments. (a) Most of the variation in $l_{gc}$ is among populations, but developmental plasticity to growth light intensity tends to increase $l_{gc}$ in sun (left facet) compared to shade (right facet) grown plants. There is no plasticity to curve type (amphi vs. pseudohypo), but the average $l_{gc}$ is slightly high in pseudohypo leaves because abaxial stomata tended to be larger than adaxial stomata on the same leaf. (b) Measurement light intensity has the most direct effect on $f_{gmax}$ because stomatal aperture increases under high light intensity. To a lesser extent, $f_{gmax}$ varies among populations and is slightly lower in sun-grown plants. ADD

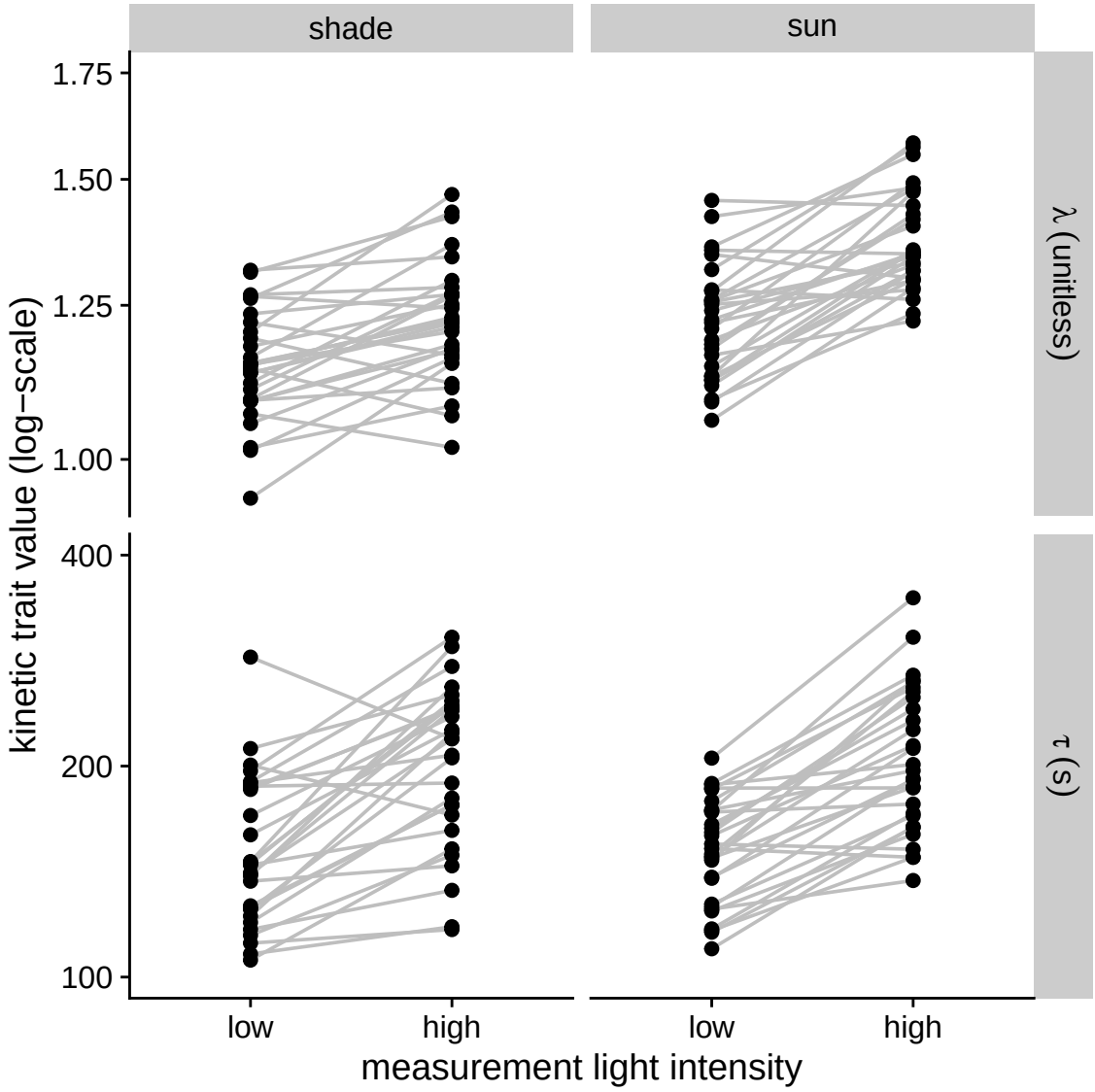


Figure 2: **Stomatal kinetics parameters $\lambda$ (lag time; top facets) and $\tau$ (time constant; lower facets) vary among wild tomato populations.** Stomatal conductance decreases faster (lower $\tau$) in response to a step change in vapor pressure deficit (VPD) in leaves when measured under low light intensity. The pattern was consistent for both shade (left facets) and sun (right facets) grown plants. The pattern for $\lambda$\$ was qualitatively similar to that for $\tau$. Each point is the average parameter value for one accession in that treatment combination. The growth and measurement light intensity treatments are described in the Materials and Methods section.

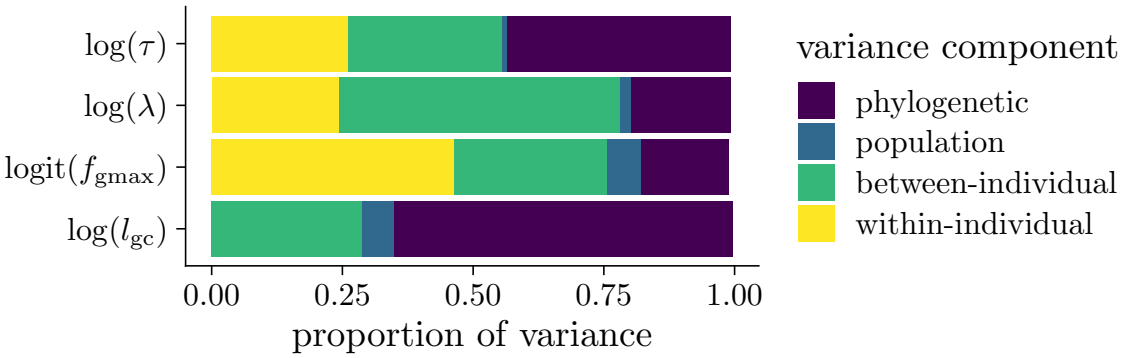


Figure 3: **foo** bar.

Table 4: Variance components of stomatal kinetic traits. The phylogenetic component is equivalent to the phylogenetic heritability. The population component is the non-phylogenetic variance among populations. Between- and within-individual variance components are among individuals within a population after accounting for treatment effect. Estimates and 95% confidence intervals (CI) are estimated as the median and quantile intervals of the posterior distribution. $f_{\text{gmax}}$: stomatal conductance as a fraction of anatomical maximum stomatal conductance; $l_{\text{gc}}$: guard cell length; $\lambda$: lag time; $\tau$: time constant.

| component | % variance | 95% CI |
|---|---|---|
| $\log(l_{\text{gc}})$ | | |
| phylogenetic | 64.7 | [32.4, 81.1] |
| population | 6.2 | [0.7, 25.6] |
| between-individual | 28.8 | [16.9, 44.2] |
| $\text{logit}(f_{\text{gmax}})$ | | |
| phylogenetic | 16.8 | [3.6, 34.7] |
| population | 6.7 | [1.9, 16.7] |
| between-individual | 29.1 | [22.9, 33.9] |
| within-individual | 46.4 | [36.4, 53.6] |
| $\log(\lambda)$ | | |
| phylogenetic | 18.9 | [5.2, 36.1] |
| population | 2.3 | [0.1, 9.1] |
| between-individual | 53.6 | [42.5, 61.6] |
| within-individual | 24.4 | [18.8, 29.8] |
| $\log(\tau)$ | | |
| phylogenetic | 42.7 | [23.3, 59.4] |
| population | 0.9 | [0.0, 8.3] |
| between-individual | 29.4 | [20.8, 38.0] |

*(continued)*

| component | % variance | 95% CI |
|---|---|---|
| within-individual | 26.1 | [18.2, 34.1] |

Guard cell size and $f_{\text{gmax}}$affected stomatal kinetics, but at different levels of biological organization. In the model with the lowest LOOIC (see Table 6 for model comparison results), the only significant partial correlation was between $l_{\text{gc}}$and $\tau$ ($\rho = 0.74$ [95% CI: $-0.12 to 0.98$)] was a significant phylogenetic correlation between $l_{\text{gc}}$and $\tau$ (**fig?-**?). PARTIAL CORRELATION RESULT. Variation in $l_{\text{gc}}$within populations did not explain within-population variation in $\tau$ Table 5. NOTE THAT BEST MODEL RETAINED INDIVIDUAL LEVEL EFFECT OF GCL, BUT COEFFICIENT WAS NOT SIGNIFICANT. OTHER BEST MODELS DID NOT RETAIN THIS. DID GCL MEDIATE AFFECT OF TREATMENTS?

Need to show (this is a growing, incomplete list): - phylogenetic partial correlation - partial correlation showing that gcl and tau are (only?) significant phylogenetic covariates

Table 5: Fixed effects (posterior mean, SE, and 95% confidence intervals).

| **Parameter** | **Estimate [95% CI]** |
|---|---|
| $\log(\lambda)$ | |
| effect of logit$(f_g)$ on $\log(\lambda)$ | 0.10 [0.08, 0.13] |
| effect of high measurement light intensity on $\log(\lambda)$ | 0.01 [-0.01, 0.03] |
| intercept (shade, low light) | 0.39 [0.28, 0.49] |
| effect of pseudohypostomatous curve type on $\log(\lambda)$ | -0.04 [-0.06, -0.03] |
| effect of sun growth treatment on $\log(\lambda)$ | 0.12 [0.10, 0.13] |
| $\log(\tau)$ | |
| effect of logit$(f_g)$ on $\log(\tau)$ | 0.19 [0.13, 0.25] |
| effect of high measurement light intensity on $\log(\tau)$ | 0.17 [0.11, 0.22] |
| intercept (shade, low light) | 5.49 [5.13, 5.84] |
| effect of pseudohypostomatous curve type on $\log(\tau)$ | -0.03 [-0.05, -0.00] |
| effect of sun growth treatment on $\log(\tau)$ | 0.05 [0.01, 0.10] |
| $\log(l_{\text{gc}})$ | |
| intercept (shade, low light) | 2.96 [2.79, 3.12] |
| effect of pseudohypostomatous curve type on $\log(l_{\text{gc}})$ | -0.06 [-0.07, -0.05] |
| effect of sun growth treatment on $\log(l_{\text{gc}})$ | 0.11 [0.11, 0.12] |
| logit$(f_g)$ | |
| effect of high measurement light intensity on logit$(f_g)$ | 0.84 [0.82, 0.87] |
| intercept (shade, low light) | -2.52 [-2.81, -2.21] |
| effect of pseudohypostomatous curve type on logit$(f_g)$ | 0.21 [0.18, 0.24] |

Table 5: Fixed effects (posterior mean, SE, and 95% confidence intervals). *(continued)*

| **Parameter** | **Estimate [95% CI]** |
|---|---|
| effect of sun growth treatment on logit($f_g$) | -0.28 [-0.35, -0.21] |

**Stomatal aperture mediates plasticity in $\tau$**

We imposed three treatments that influenced stomatal kinetics: growth light intensity, measurement light intensity, and curve type. All three treatments directly effect $\tau$ and had indirect effects mediated by $f_{\text{gmax}}$ (Figure 4). We did not estimate effects mediated by $l_{\text{gc}}$ because the selected model did not retain direct effects of $l_{\text{gc}}$ on $\tau$ (Table 6). The direct effect of the high light intensity (sun) growth treatment increased $\tau$ by $5.64\%$ [95% CI: $1.03$ to $10.55\%$], but the indirect of lower $f_{\text{gmax}}$ counteracted by $-5.01\%$ [95% CI: $-7.14$ to $-3.12\%$]. In contrast, direct and indirect effects of high measurement light intensity increased $\tau$. The direct effect increased $\tau$ by $18.28\%$ [95% CI: $11.85$ to $24.98\%$] and the indirect effect mediated by $f_{\text{gmax}}$ increased $\tau$ by $16.83\%$ [95% CI: $11.04$ to $23.14\%$]. The direct effect of the pseudohypo treatment decreased $\tau$ slightly by $-2.60\%$ [95% CI: $-4.93$ to $-0.18\%$], but the indirect effect mediated by greater $f_{\text{gmax}}$ increased $\tau$ by $3.98\%$ [95% CI: $2.60$ to $5.50\%$].

## Discussion

effect of fgmax is not simply because high light intensity increased fgmax and tau. Causal inference dissects different causal paths and shows direct effect of fgmax on tau within populations and treatments

caveat: this treatment is a hammer than exposes maximum rate of stomatal closure. The factors which influence this may not explain variance in kinetics in more natural conditions.

Need to mention subsaturation. If airspace is unsaturated, then gs is underestimated. This doesn't necessarily affect kinetic parameter estimates since all measurements were done at high VPD.

(not sure where this goes) It might seem that natural selection should favor faster stomatal responses to get closer to the ideal of instantaneous optimization. Under this assumption, variation in stomatal kinetics would depend on how strong directional selection for faster responses is counterbalanced by costs of faster stomata. We discuss multiple, nonmutually exclusive costs below [WHERE?]. However, once models account for inevitable time lags, natural selection can favor slower responses when this prevents overreacting to unpredictable and shortlived changes in the environment. [Review some lit] It is therefore likely that the direction of selection on stomatal response rate varies depending on the predictability and duration of environmental fluctuations, as well as the energetic costs of stomatal movements.

Traditional view is that smaller stomata are selected for speed and because there is higher gmax for same fs (Franks). Our results suggest an alternative hypothesis. Smaller stomata may increase rate of

Table 6: Model comparison results. Models with a $\Delta$LOOIC less than two standard errors (SE) from the minimum LOOIC are considered to have substantial support and are used for posterior predictions to test hypotheses about the effects of guard cell length ($l_{gc}$) and the stomatal conductance as a fraction of anatomical maximum stomatal conductance ($f_{gmax}$) on stomatal kinetic parameter $\lambda$ (time lag) and $\tau$ (time constant). LOOIC stands for leave-one-out cross-validation information criterion.

| | | $\lambda$ | | $\tau$ | |
|---|---|---|---|---|---|
| $\Delta$LOOIC | SE | $f_{gmax}$ | $l_{gc}$ | $f_{gmax}$ | $l_{gc}$ |
| 0.00 | 0.00 | ✓ | | ✓ | |
| 2.60 | 2.55 | ✓ | ✓ | ✓ | |
| 3.22 | 2.62 | ✓ | | ✓ | ✓ |
| 3.55 | 2.71 | ✓ | ✓ | ✓ | ✓ |
| 17.15 | 10.63 | ✓ | ✓ | | ✓ |
| 18.30 | 10.51 | ✓ | | | ✓ |
| 20.75 | 10.47 | ✓ | ✓ | | |
| 23.97 | 10.56 | ✓ | | | |
| 46.10 | 14.17 | | ✓ | ✓ | ✓ |
| 46.41 | 14.14 | | | ✓ | |
| 49.51 | 14.17 | | ✓ | ✓ | |
| 49.95 | 14.16 | | | ✓ | ✓ |
| 68.29 | 17.98 | | | | ✓ |
| 68.83 | 18.00 | | ✓ | | |
| 69.85 | 18.00 | | ✓ | | ✓ |
| 71.45 | 17.94 | | | | |

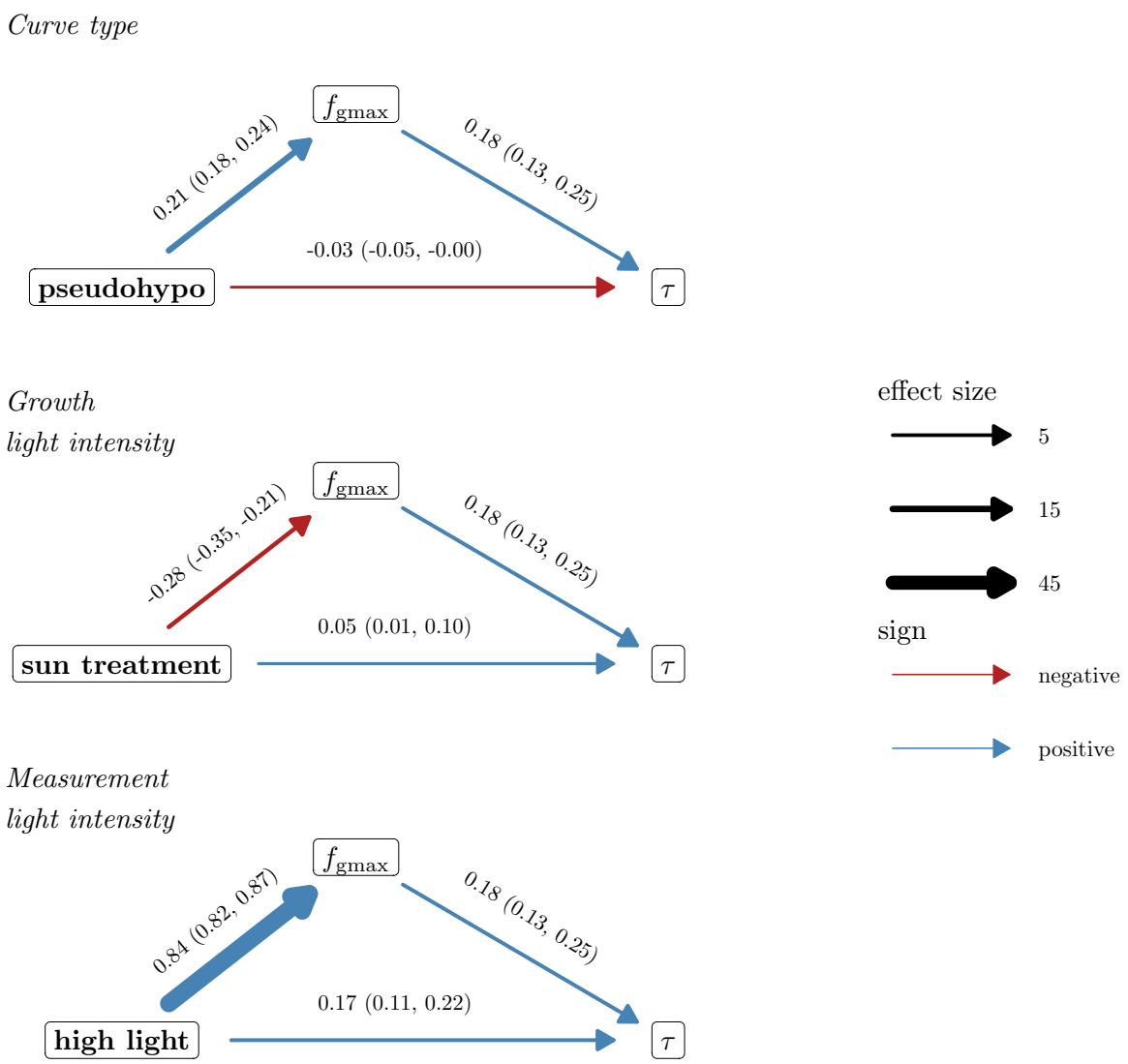


Figure 4: **Stomatal aperture as a fraction of anatomical maximum stomatal conductance ($f_{gmax}$) mediates plasticity in the time constant $\tau$.** The three treatments are curve type (top facet; amphi vs. pseudohypo), growth light intensity (middle facet; shade vs. sun), and measurement light intensity (bottom facet; low vs. high). Horizontal arrows at the bottom of each graph indicate the direct effect of the treatment level on $\tau$; slanted arrows leading to or from $f_{gmax}$ indicate mediated effects of the treatment on $\tau$. Line thickness is proportional to the effect size and line color indicates whether the effects are negative (red) positive (blue). Parameter estimates of the effect and 95% confidence intervals are above each line. Note that are not consistent because $f_{gmax}$ is on a logit scale whereas $\tau$ is on a log-seconds scale. The selected model did not retain direct effeects of $l_{gc}$ on $\tau$ and hence this variable could not be considered as a mediator.

stomatal closure (we haven't tested opening), but only if fgmax is low or moderate. Therefore if leaves with smaller stomata maintain a similar gop, then stomatal density must increase to prevent fgmax from increasing. This may explain the association between the association between small stomata and high density, and relatedly the dearth of leaves with sparse, small stomata.

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# Supplementary Materials

## Supporting figures

**Figure file:**

Download from: https://github.com/cdmuir/solanum-kinetics/raw/refs/heads/main/figures/rh-curves.pdf

Figure S1: : Humidity-response curves and fitted lines. The title provides the Tomato Genetics Resource Center accession number, replicate letter, and species names. The subtitle indicates the growth light intensity (sun or shade), measurement light intensity (low or high), and curve type (amphi or pseudohypo). Points are raw data and lines are fitted curves. The Bayesian correlation coefficient, Bayes $R^2$, is shown to the right of each curve.

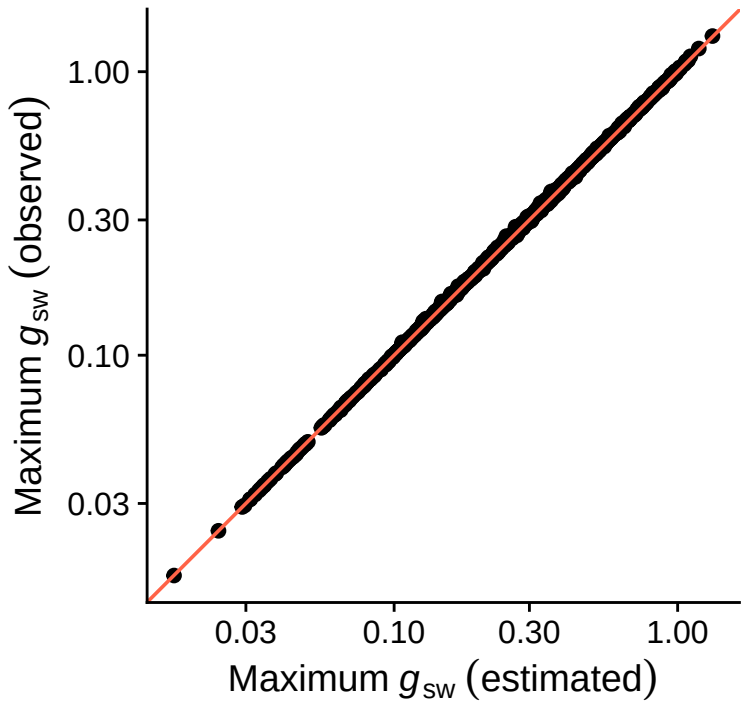


Figure S2: **foo** bar.

## MANUSCRAPS

Note: this theory did not really pan out. It did not come up with models that really fit better than the CD Weibull model and they included an extra parameter. Therefore, I am tabling this for now and trying OnGuard3 instead.

**Theory**

Our goal here is derive equations that integrate the effects of guard cell size and aperture on stomatal kinetics. A fully mechanistic model is beyond the scope of our goals here. Instead, we use phenomenological equations that adequately capture empirical patterns. There are three main assumptions of our model. First, we assume linear relaxation kinetics for guard cell turgor pressure ($P$):

$$\frac{dP}{dt} = \frac{P^* - P}{\tau}.$$

In this differential equation, $P^*$ is the equilibrium turgor pressure and $\tau$ is a time constant that determines the speed of relaxation. Solving this equation gives:

$$P(t) = P^* + [P(0) - P^*]e^{-t/\tau},$$

where $P(0)$ is the initial turgor pressure at time $t = 0$.

The rate of relaxation is determined by the surface area to volume ratio. Assume constant transport per surface area. Surface area determined by geometry.

What is eqn for this?